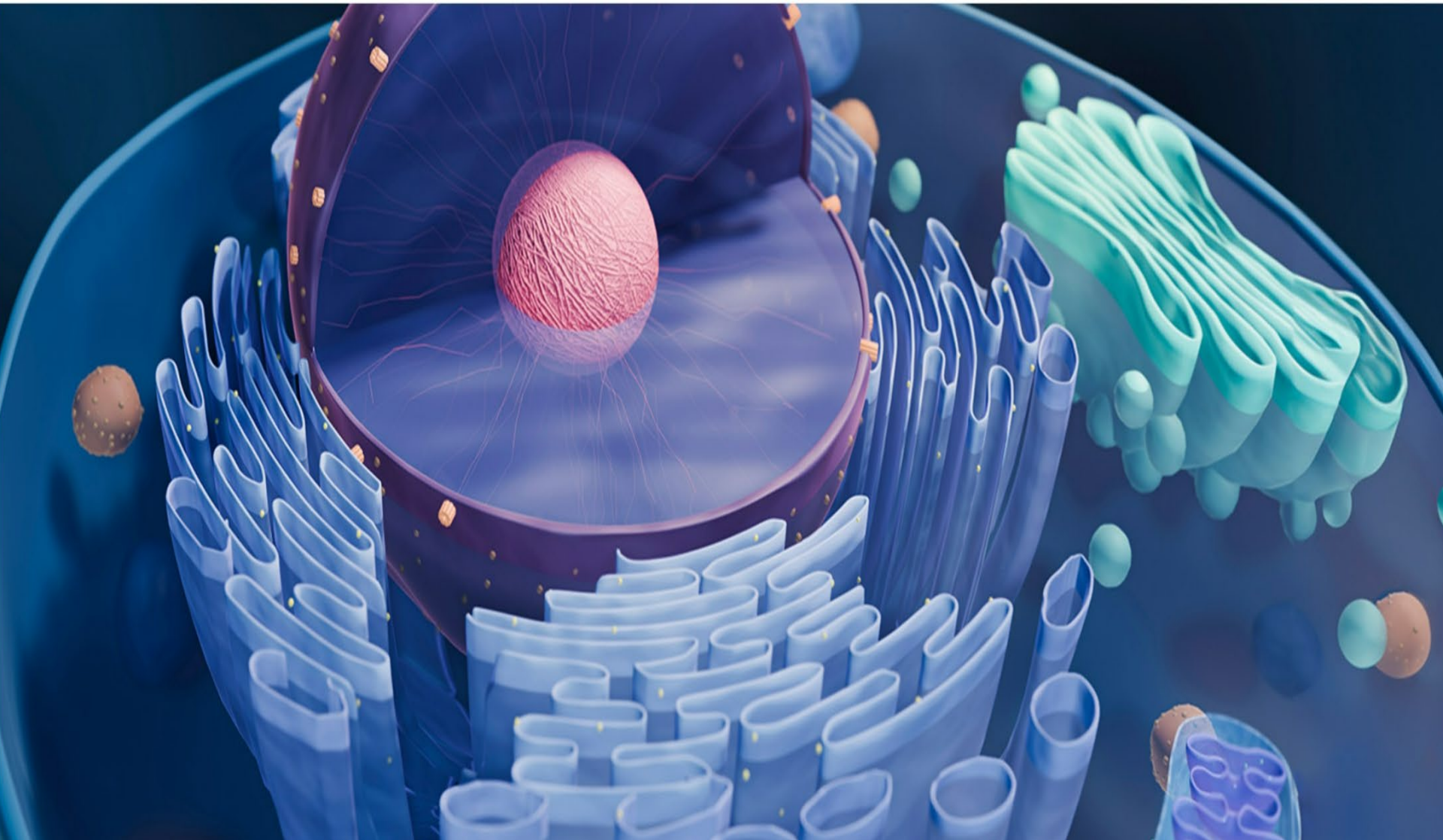


The Endomembrane System



- The endomembrane system is the set of cytoplasmic cavities limited by membranes inter-communicating with each other through vesicles or canaliculi.
- These organelles exist only in eukaryotic cells.
- The different compartments of this system are:

The endoplasmic reticulum

The Golgi apparatus

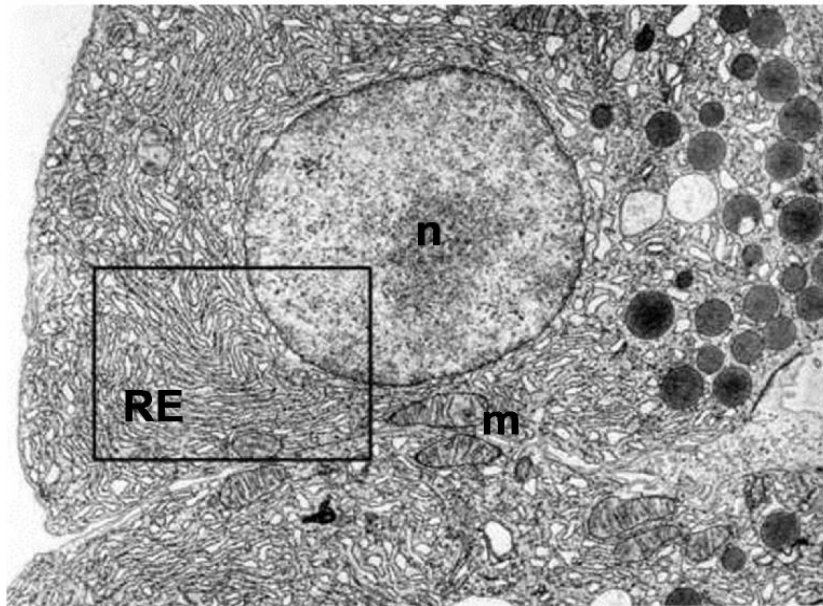
Endosomes

Lysosomes

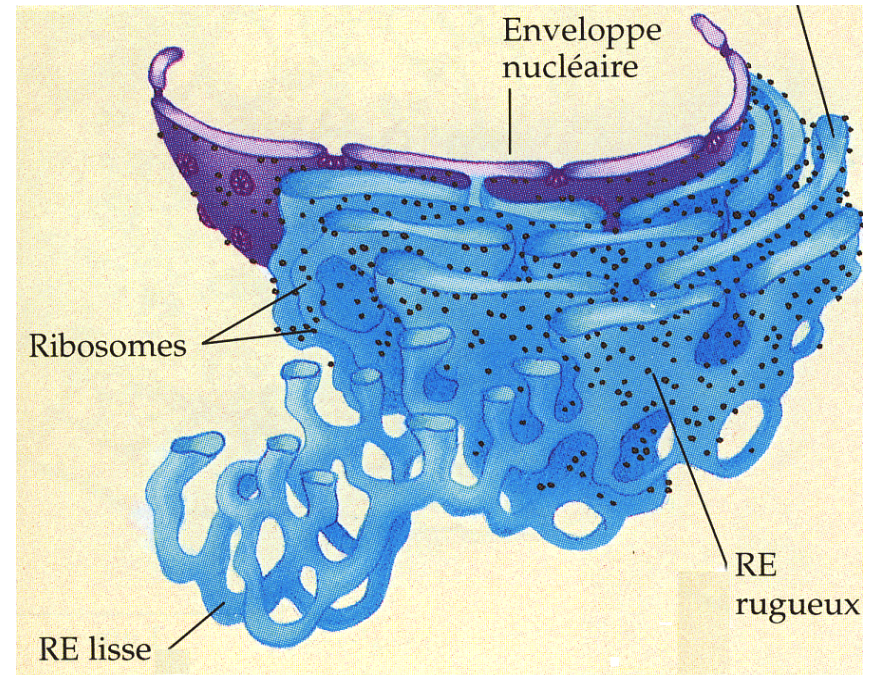
Nb: mitochondria and peroxisomes are not part of this system

The endoplasmic reticulum

Is a predominant organelle occupying about 20% of the cell volume in eukaryotes. Isolated from the cytosol by a membrane that forms a continuous sheet delimiting an internal space "light or cistern" and creating a network of cavities or more or less dilated vesicles communicating with each other. Observation under an electron microscope shows continuity between the reticular membrane and the nuclear envelope.



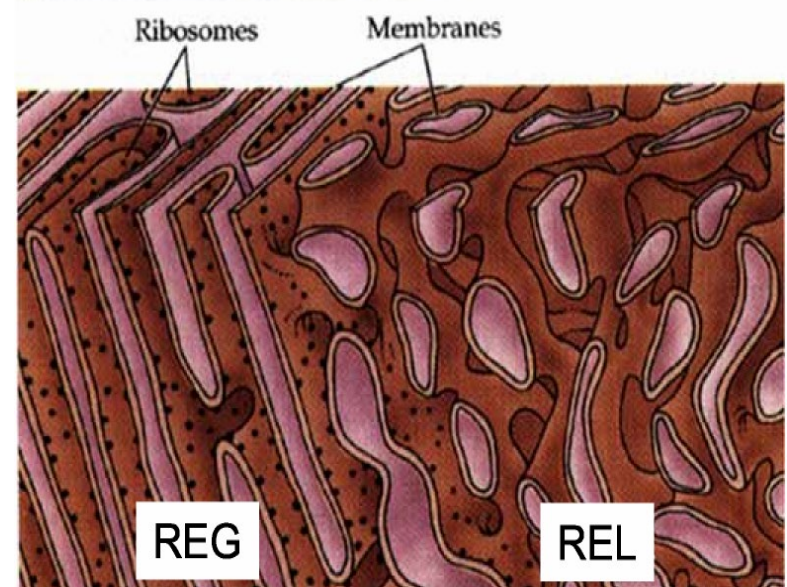
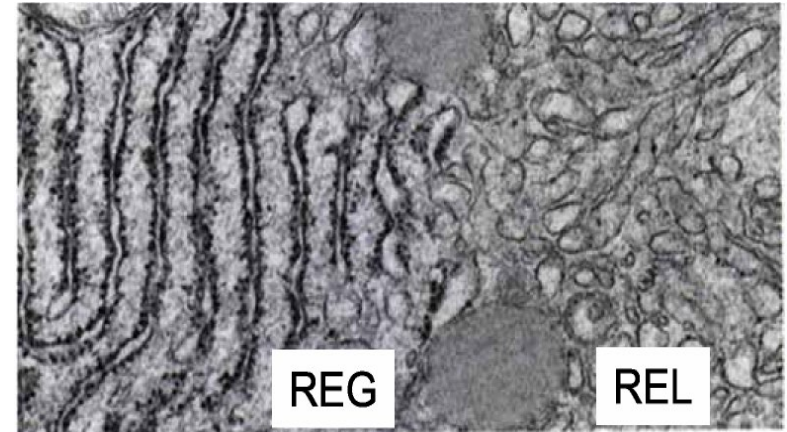
Cell observation with an electron microscope



2 types of reticulum are defined:

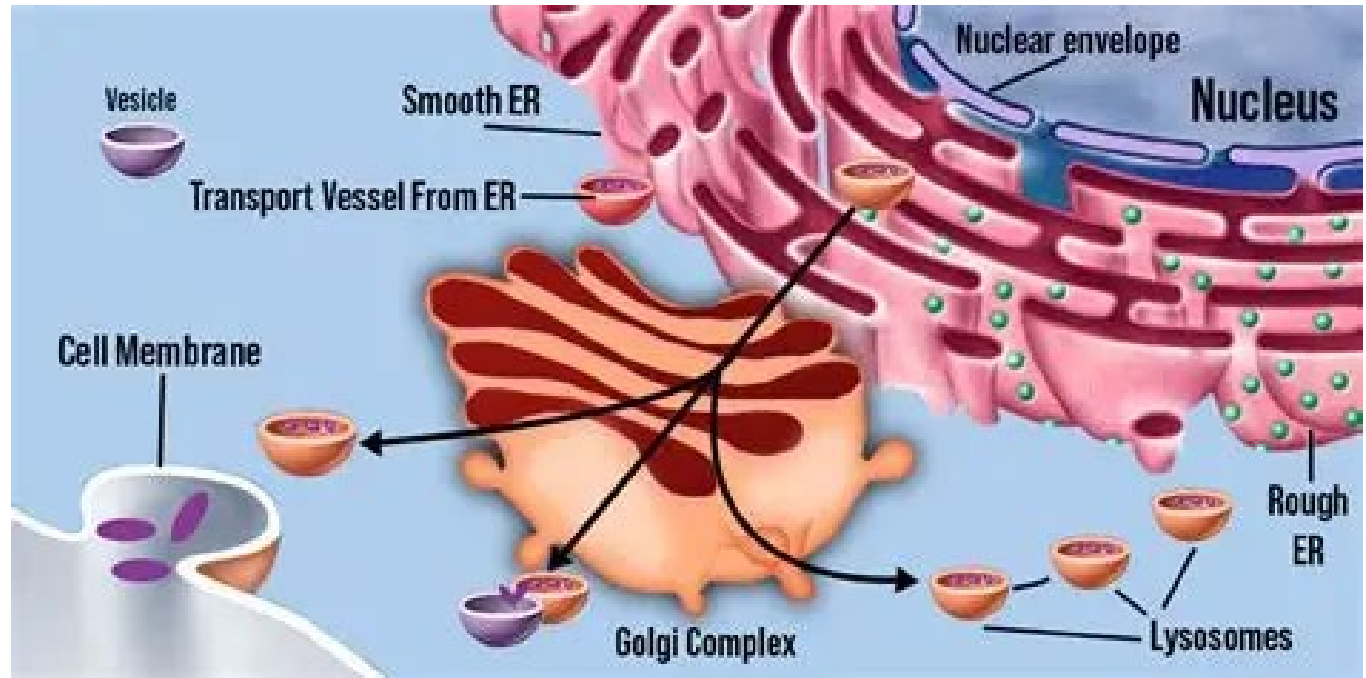
The endoplasmic reticulum smooth (REL)

The endoplasmic reticulum granular (REG)
which features ribosomes



PERCENTAGE OF CELL MEMBRANES

REG and REL can coexist in the same cell and can communicate with each other (liver cell). In the case of active protein synthesis, the REG is usually highly developed (pancreatic cells). In the case of high lipid metabolism, REL is very abundant (liver cell).

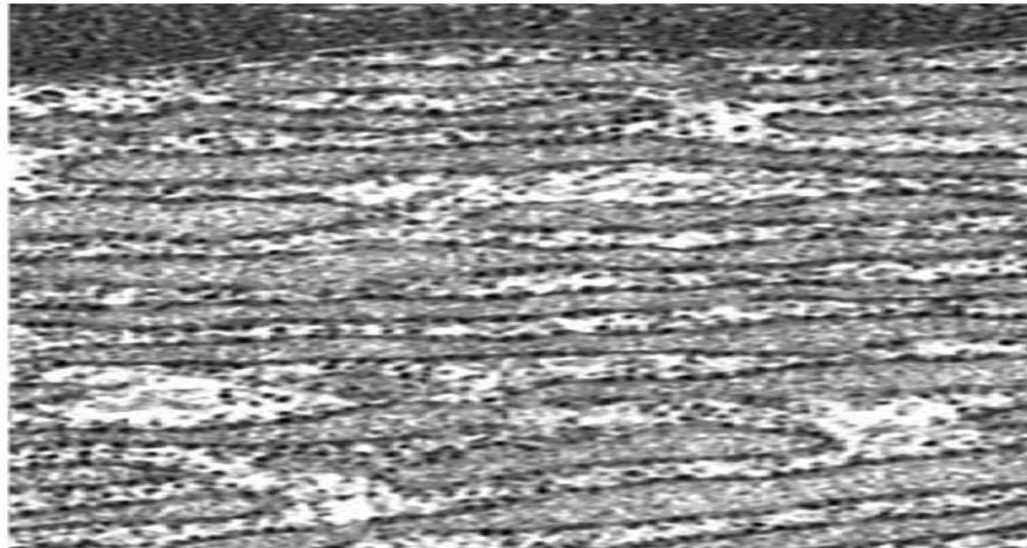


The chemical composition of the endoplasmic reticulum

Biochemical studies performed on the reticular membrane after isolation by centrifugation have shown that it contains (70%) proteins, usually enzymes, and (30%) lipids.

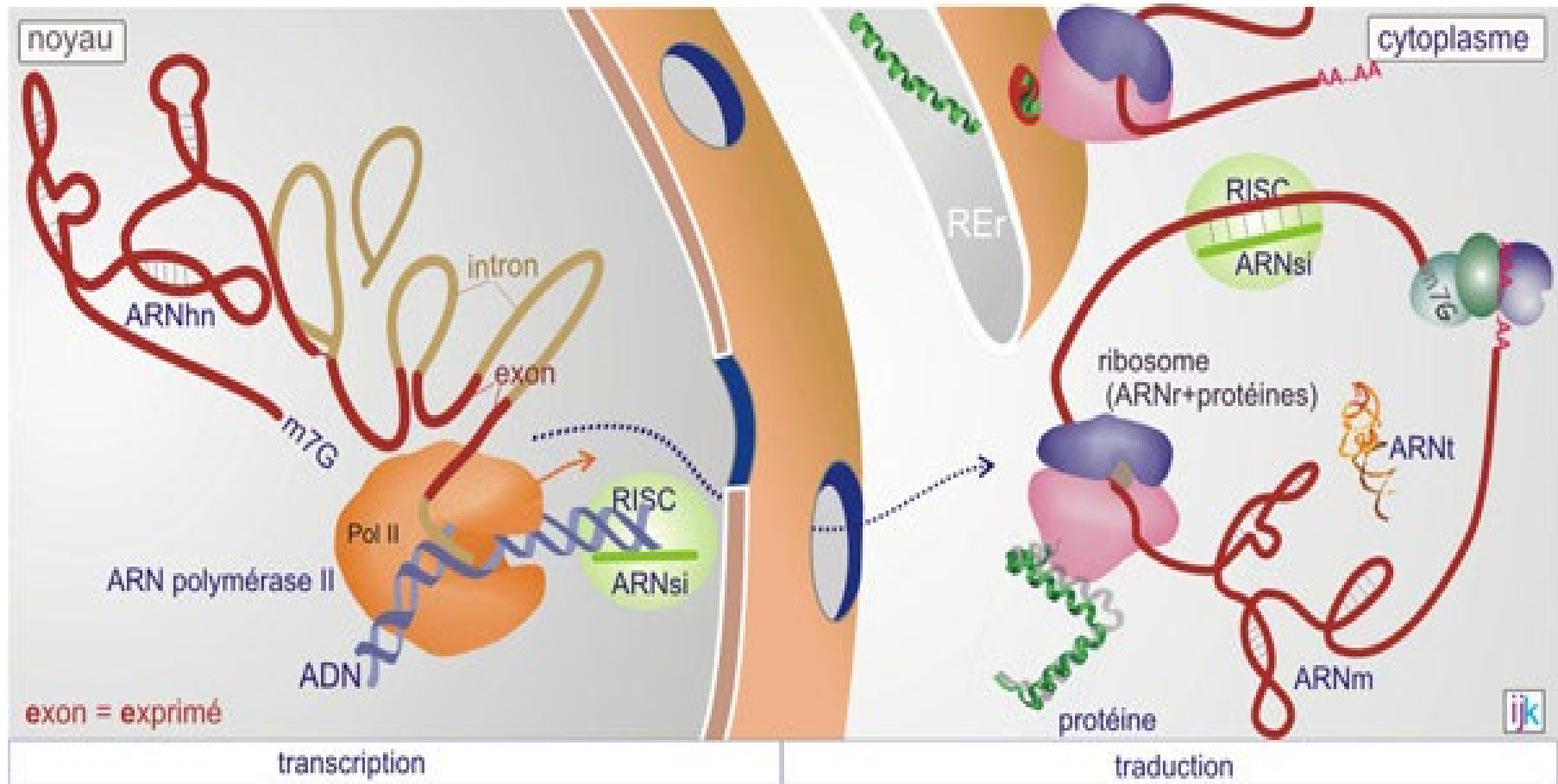
The reticular membrane presents a very marked asymmetry since on the cytosolic side we find: cytochrome b₅, cytochrome P450 reductase and the Ca⁺⁺ - dependent ATPase.

On the luminal side we have: glucose-6-phosphatase, β glucuronidase and glycosyl transferases.



Role of ERG in protein synthesis

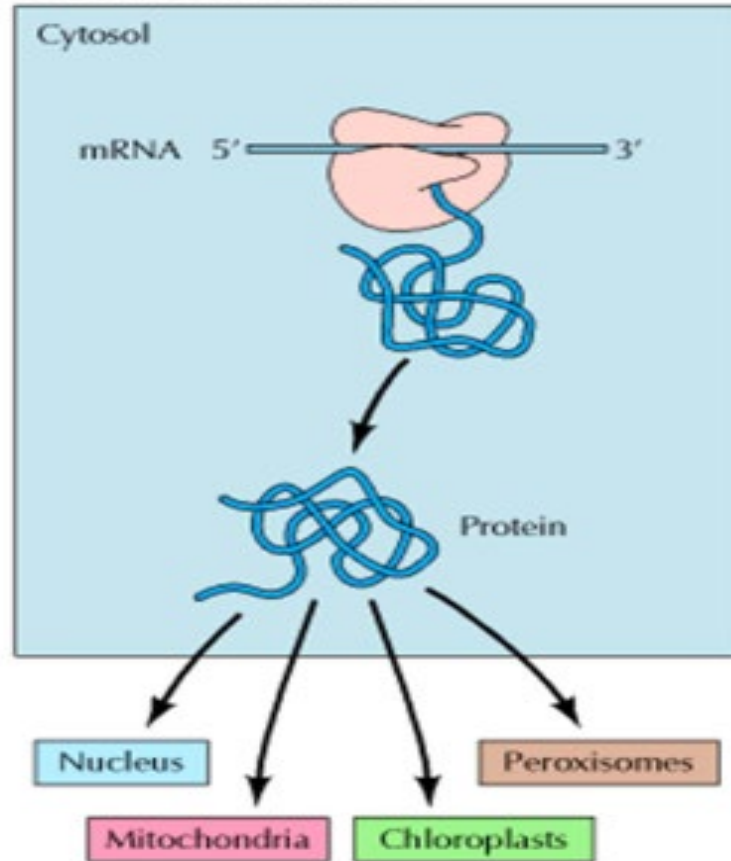
ADN $\begin{cases} \rightarrow \text{ARNmi} \\ \rightarrow \text{ARNm} \end{cases}$ $\rightarrow \text{ARNsi} \rightarrow \text{protéine}$



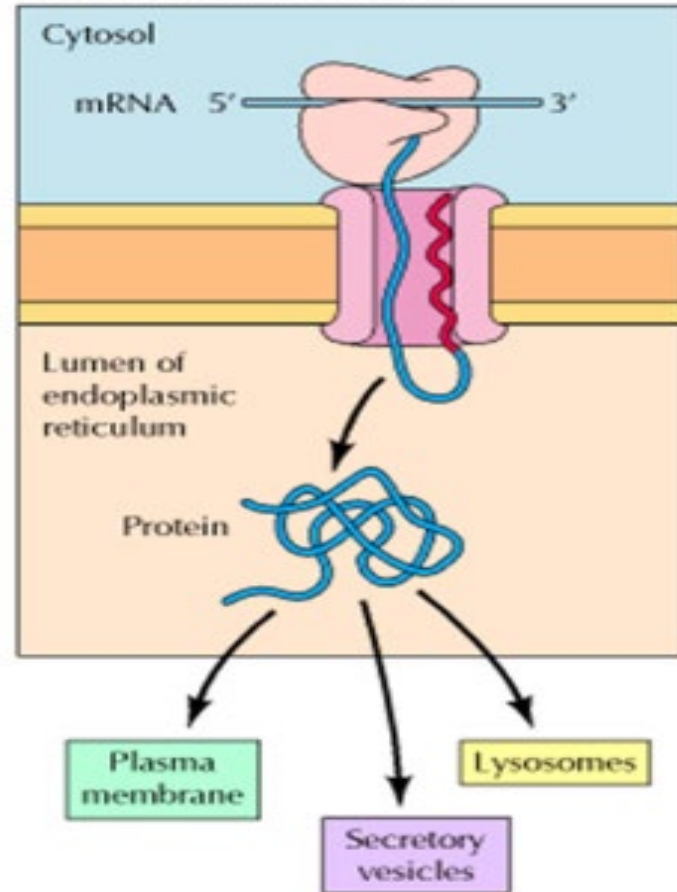
Proteins Synthesis

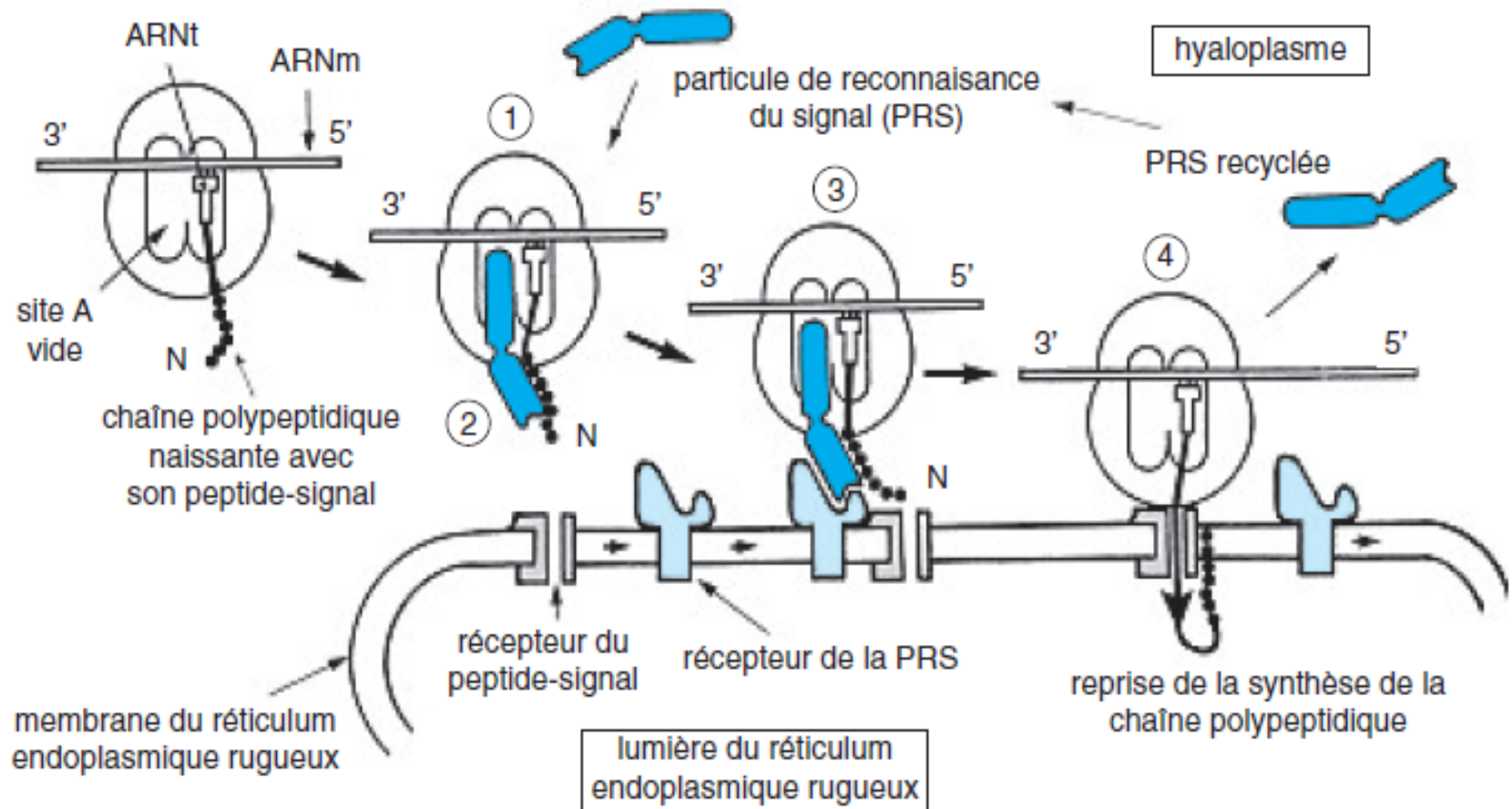
There are two different types of targeting for newly synthesized proteins

Free ribosomes in cytosol



Membrane-bound ribosomes

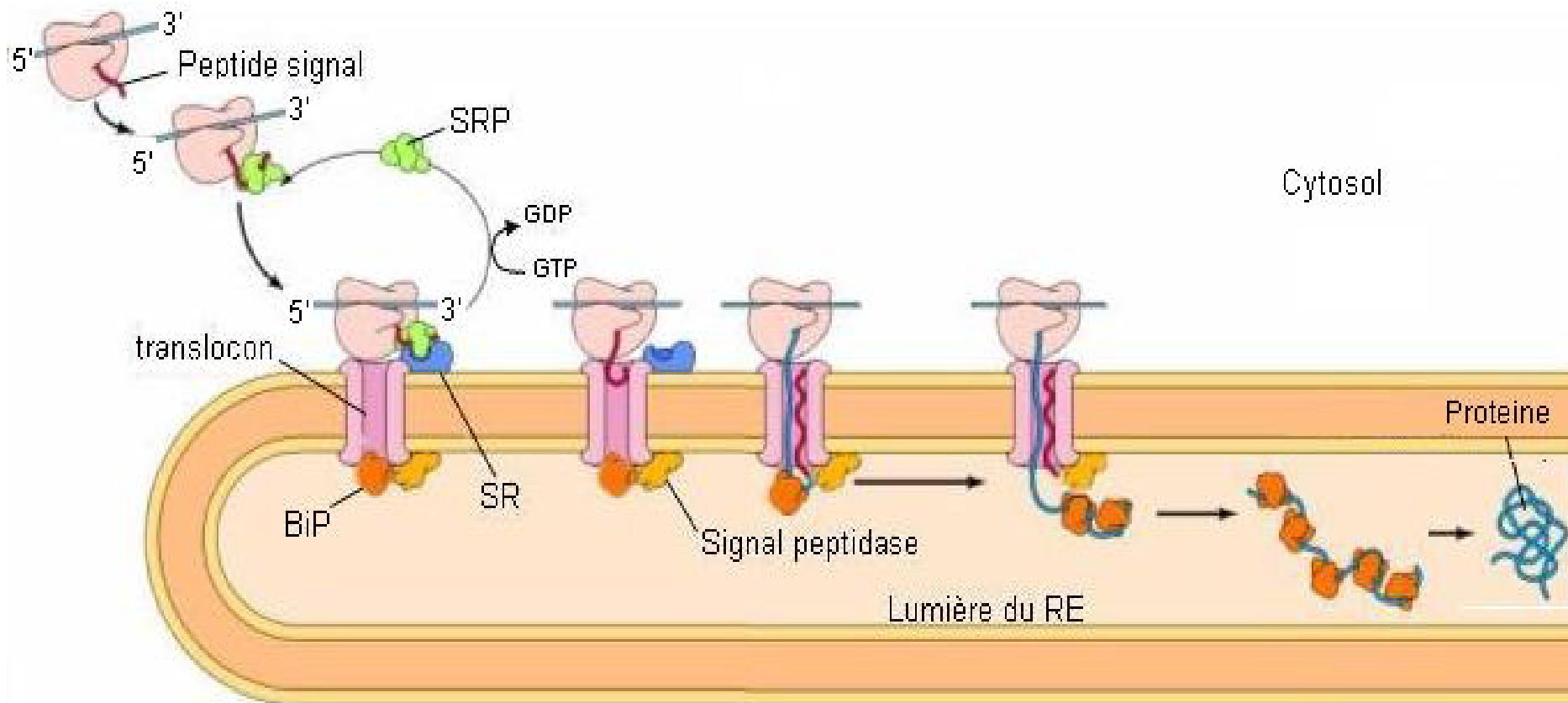




Mechanism of Co Translational Insertion of Proteins into the Rough Endoplasmic Reticulum

The role of PRS as a targeting agent and that of the various membrane receptors are highlighted during the different stages of this process.

- (1) Recognition of the signal sequence and binding of PRS.
- (2) Complex formed by the ribosome and PRS, located in the A site and blocking translation.
- (3) Attachment of the complex to the RR membrane.
- (4) Resumption of translation and injection of the polypeptide chain through the membrane.



Translocation co-translationnelle des protéines (Membranes plasmiques, Lysosomes, Vésicules sécrétoires)

Molecular Biology of the Cell

Protein synthesis begins at the nucleus, where the transcription phase takes place, following which an mRNA is released into the cytosol. Of all the ribosomes, only some having begun the synthesis of proteins in a free state in the hyaloplasma, are directed to the membranes of the reticulum; From this point on, the fate of the proteins made by these ribosomes is fixed and only a small number of compartments will be accessible to them.

The second step begins with the attachment of the mRNA by its 5' end at its site on the free ribosome, this mRNA has on its 5' end after the AUG codon a special nucleotide sequence which, once translated, gives the formation of the hydrophobic residue peptide called the signal peptide.

An active compound called the signal recognition particle (SRP) attaches to the N-terminal part of the signal peptide by blocking protein synthesis, the SRP-signal peptide complex makes its way to the membrane of the ER where it attaches to a receptor of the SRP. After contact, the SRP is dissociated from the complex and released into the cytosol.

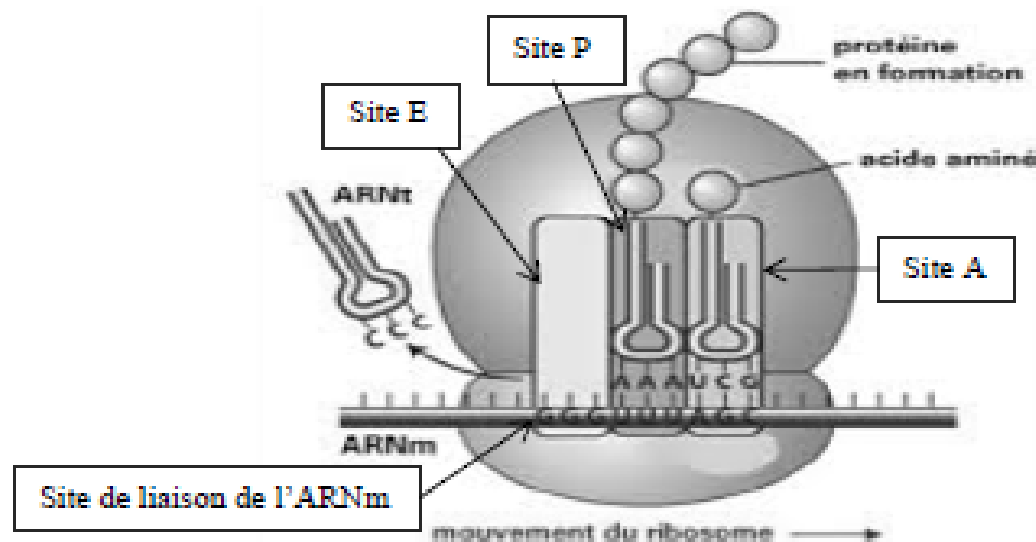
Protein synthesis resumes and the protein being synthesized penetrates and translocates through the double lipid layer of the ribosome.

Once the protein passes through the port; peptidases cut the signal peptide of the protein, then it is degraded by ER proteases. In the end, the synthesized protein is released into the lumen of the ER, where it acquires its three-dimensional shape.

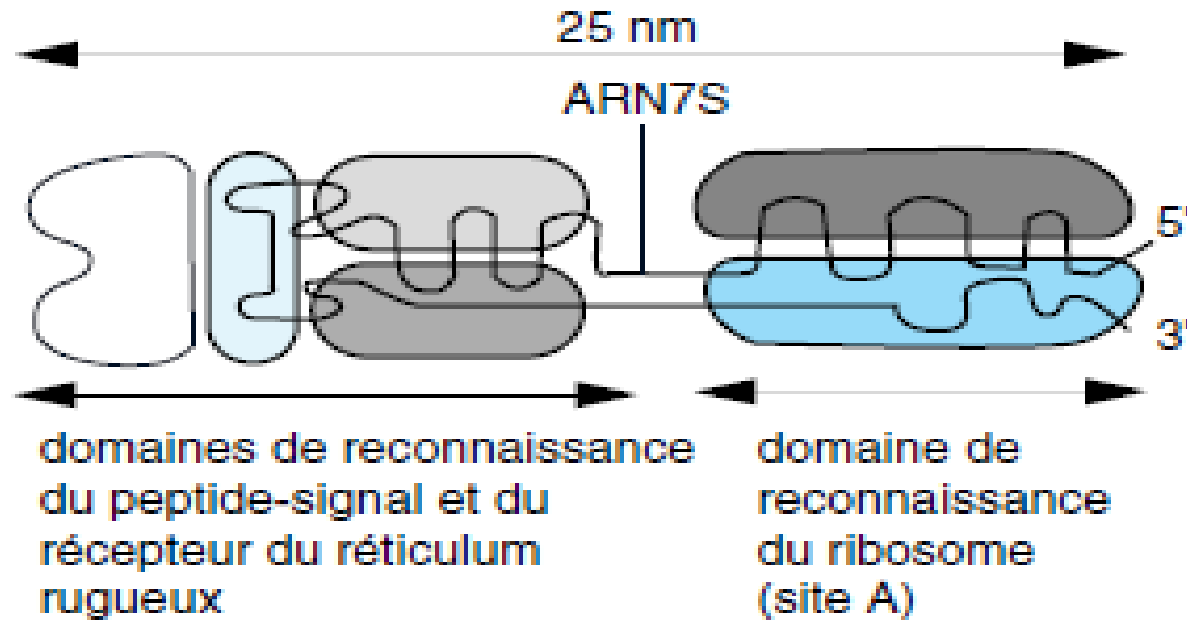
The ribosomes

The ribosome has four binding sites located exclusively on rRNAs: one mRNA binding site located on the rRNA of the small subunit and three ATP binding sites located primarily on the rRNA of the large subunit.

- The aminoacyl-tRNA binding site (A site), which binds the incoming tRNA molecule carrying a new amino acid.
- The peptidyl-tRNA binding site (P site), which binds the tARN molecule carrying the growing polypeptide; this is the site where a new peptide bond forms between two amino acids. This site is formed by 23S rARN. Peptidyltransferase is therefore not a ribosomal protein enzyme but an rRNA acting as an enzyme (hence the name ribozyme for the ribosome).
- And finally, the binding site of the outgoing empty tRNA (E or Exit site).



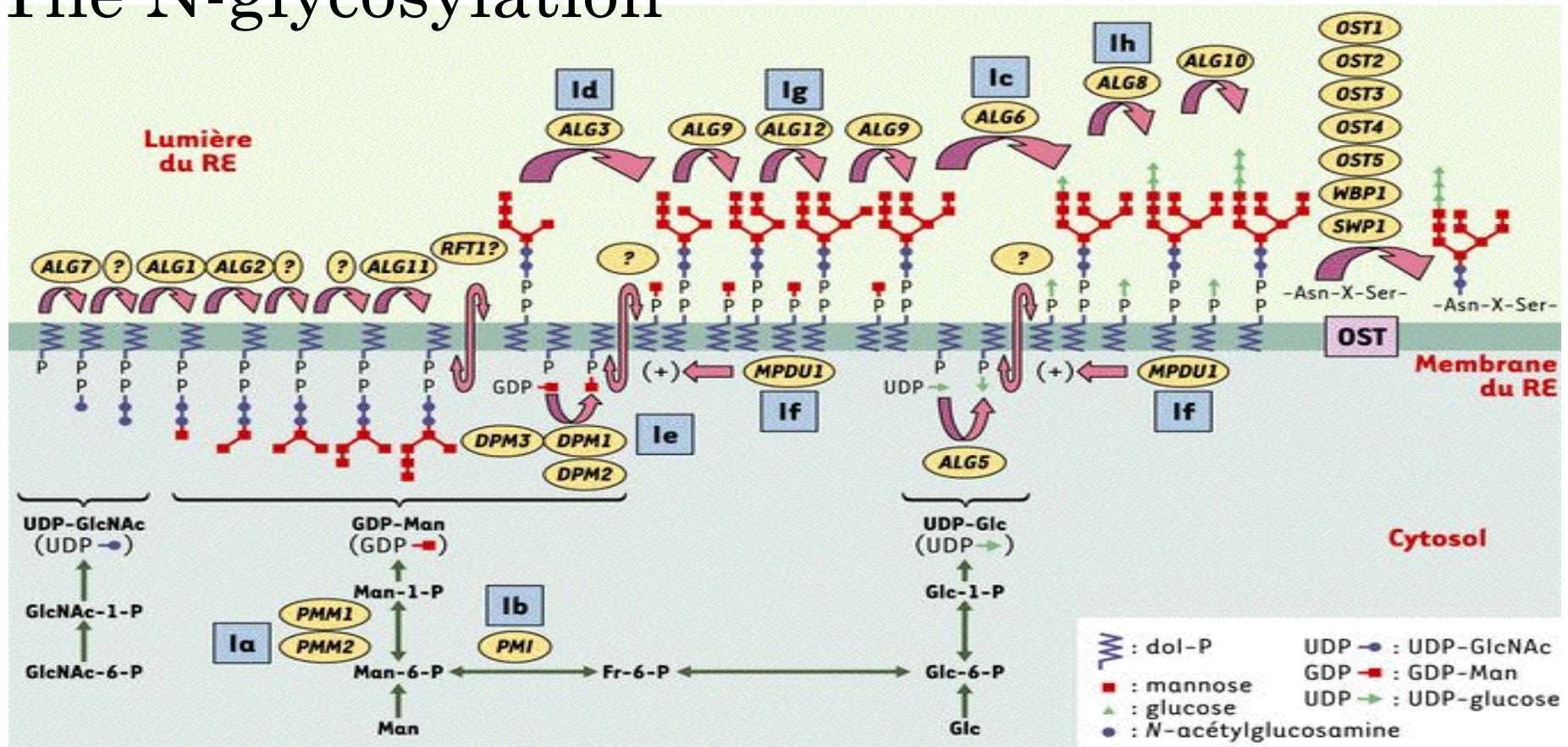
The signal recognition particle (SRP)



This large particle (25 nm) is a ribonucleoprotein (1 ARIN of 300 nucleotides and 6 polypeptide chains). It possesses three receptors, through which it can recognize:

- 1) the amino acid site of ribosome
- 2) the signal sequence of a growing protein
- 3) the receptor carried by the Reticulum.

The N-glycosylation



Polysaccharides attached via an N-glycosidic bond come from a precursor molecule containing 14 oses: N-acetylglucosamine (GlcNAc), mannose (Man) and glucose (Glc).

These oses are successively added to dolichol, an isoprenoid lipid embedded in the membrane of the endoplasmic reticulum.

The first 7 oses are brought by sugar-nucleotides (UDP-osés and GDP-osés) into the cytoplasm and attached to dolichol via a pyrophosphate (PP) bond.

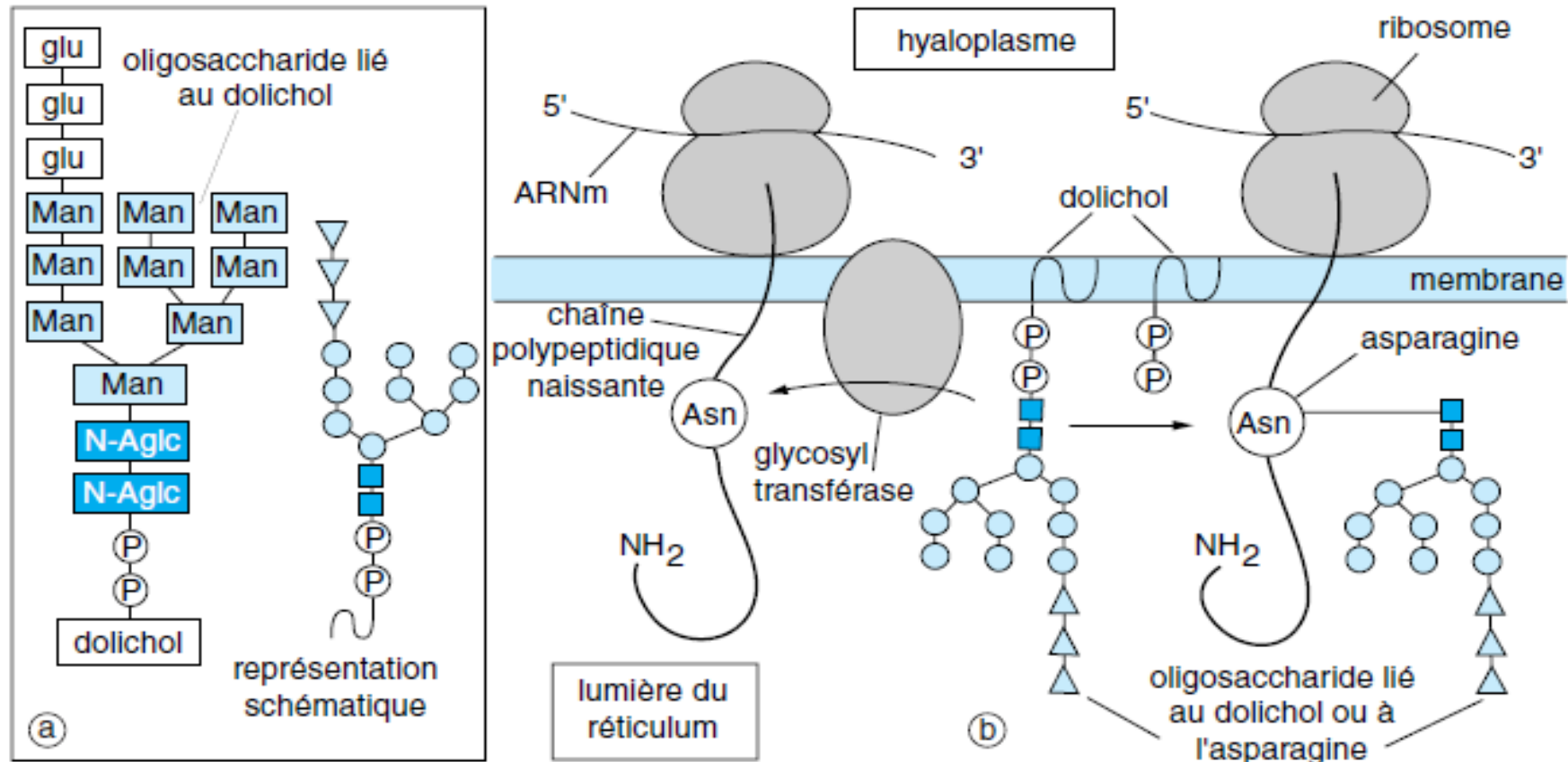
After the Man₅GlcNAc₂-PP-dolichol intermediate is formed, the entire complex switches into the lumen of the endoplasmic reticulum.

The 7 terminal oses are provided by Man-P-dolichol and Glc-P-dolichol molecules to form the precursor glycan Gcl₃Man₉GlcNAc₂-PP-dolichol.

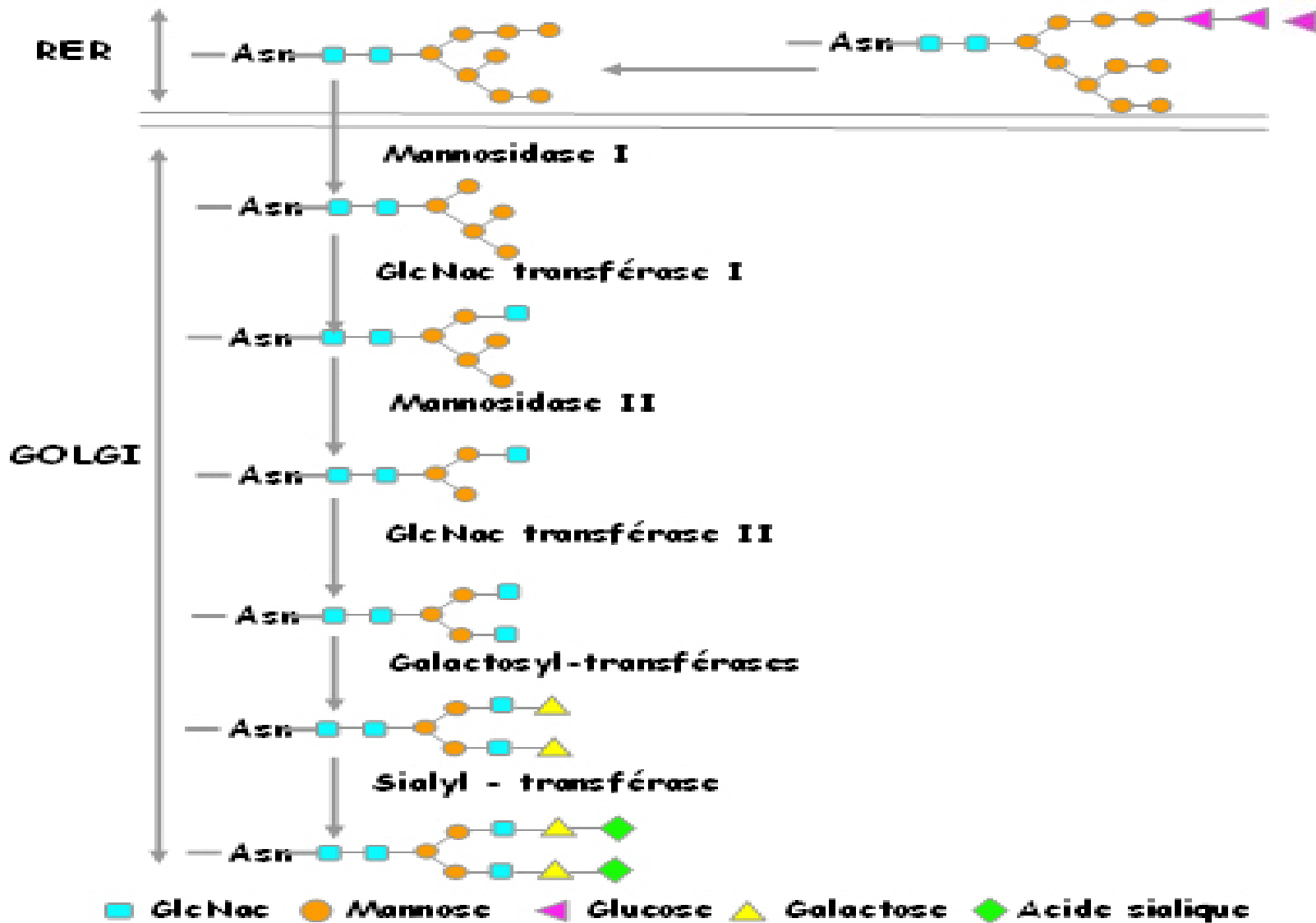
N-glycosylation is essentially co-translational: in fact, glycan is attached to the polypeptide chain during biosynthesis and during its transport in the endoplasmic reticulum.

The N means that the glycan is covalently attached to the nitrogen of the amide group of an asparagine (Asn-X-[Ser/Thr]) pattern).

The sugar immediately attached to asparagine is N-acetylglucosamine.

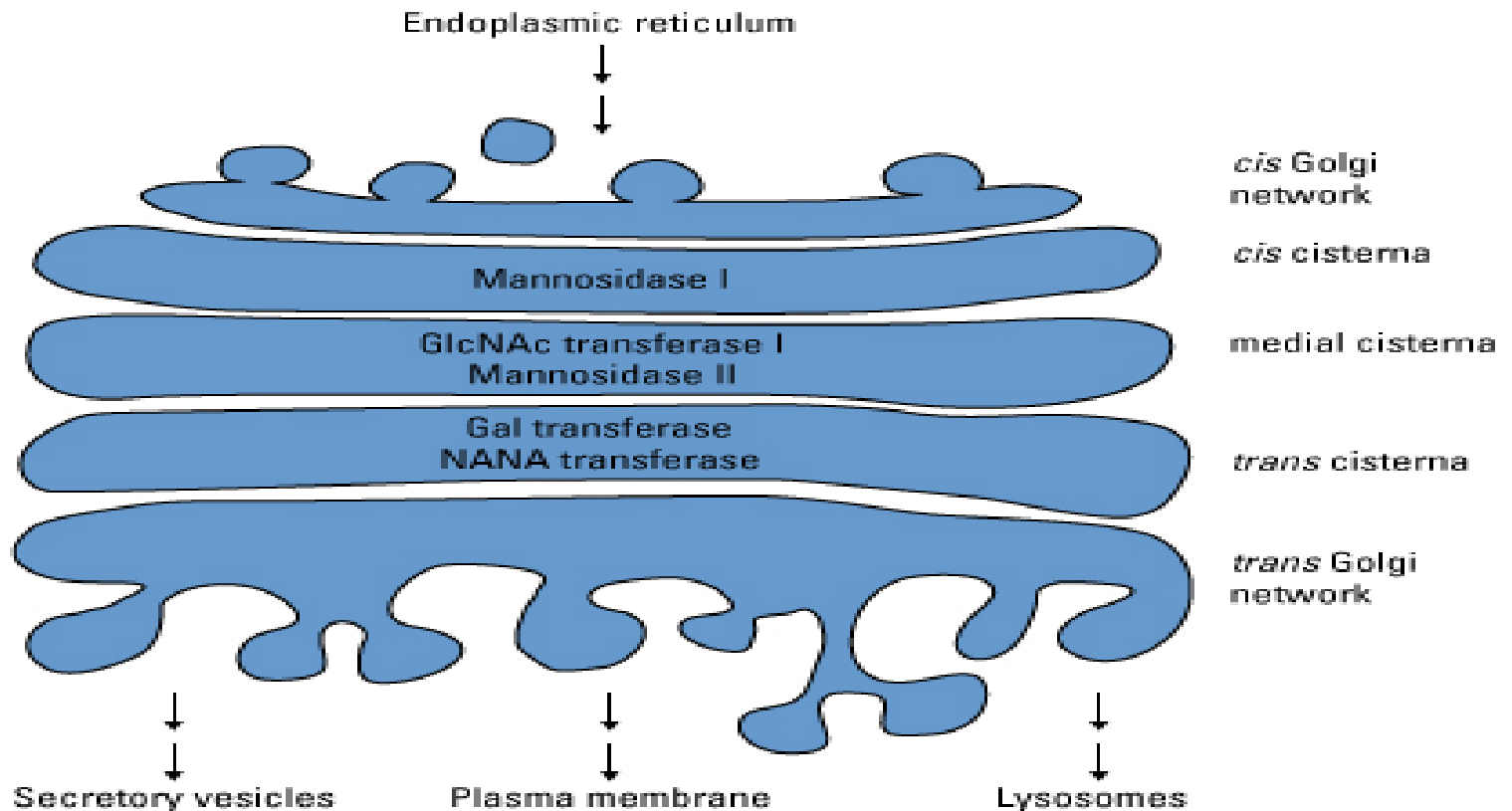


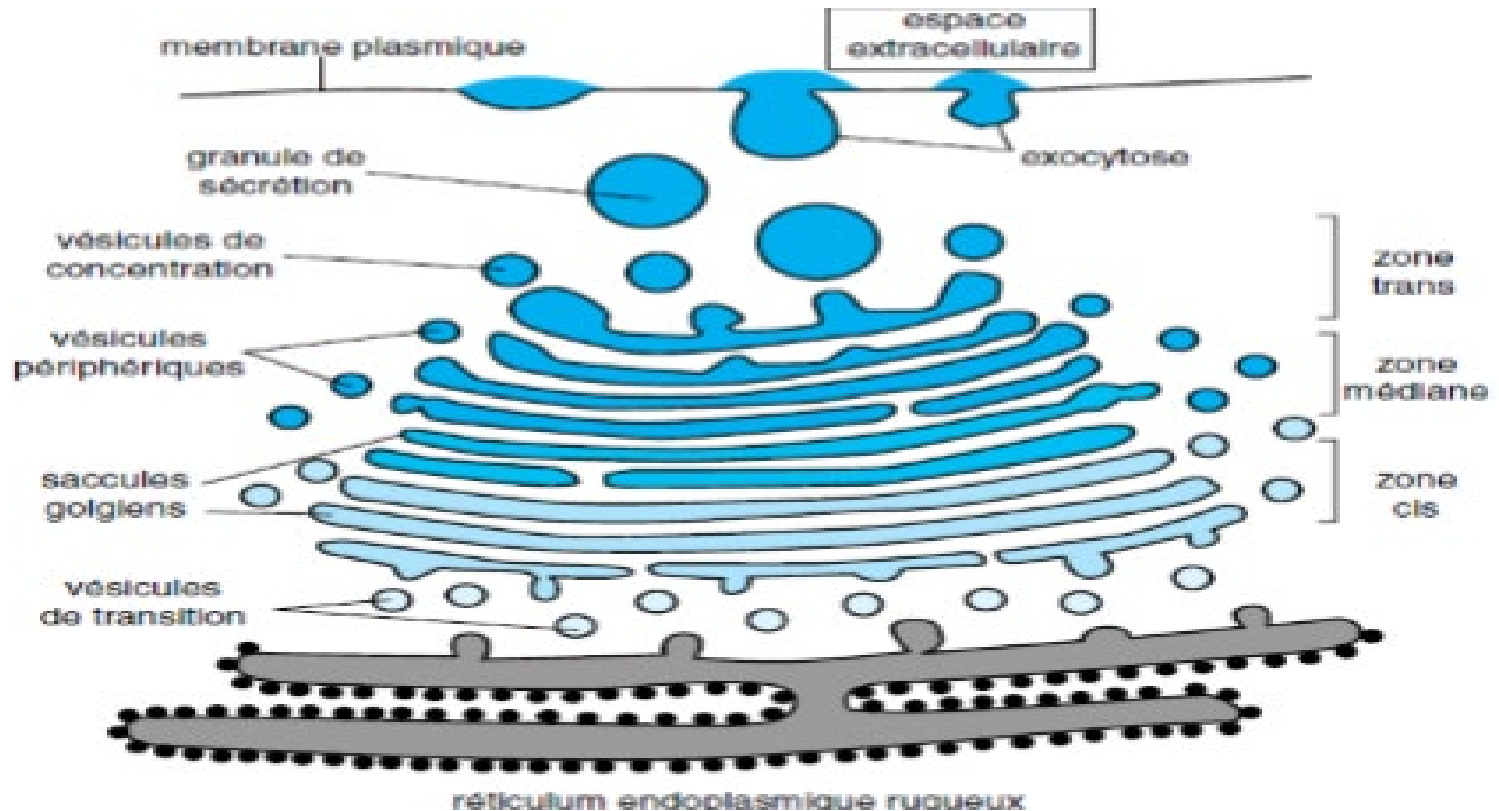
N-glycosylation of proteins in the Golgi



The Golgi Apparatus

The discovery of this organelle was made by the Italian biologist Camillo GOLGI (Nobel Prize 1906). It consists of a set of parallel flattened cisterns or saccules associated with secretory vesicles. The saccules of a dictyosome are 4 to 6 in number, and there may be a connection between the lumens of two neighboring saccules. The Golgi apparatus is highly developed in cells specialized in secretion, as is the case in cells of the intestinal epithelium.



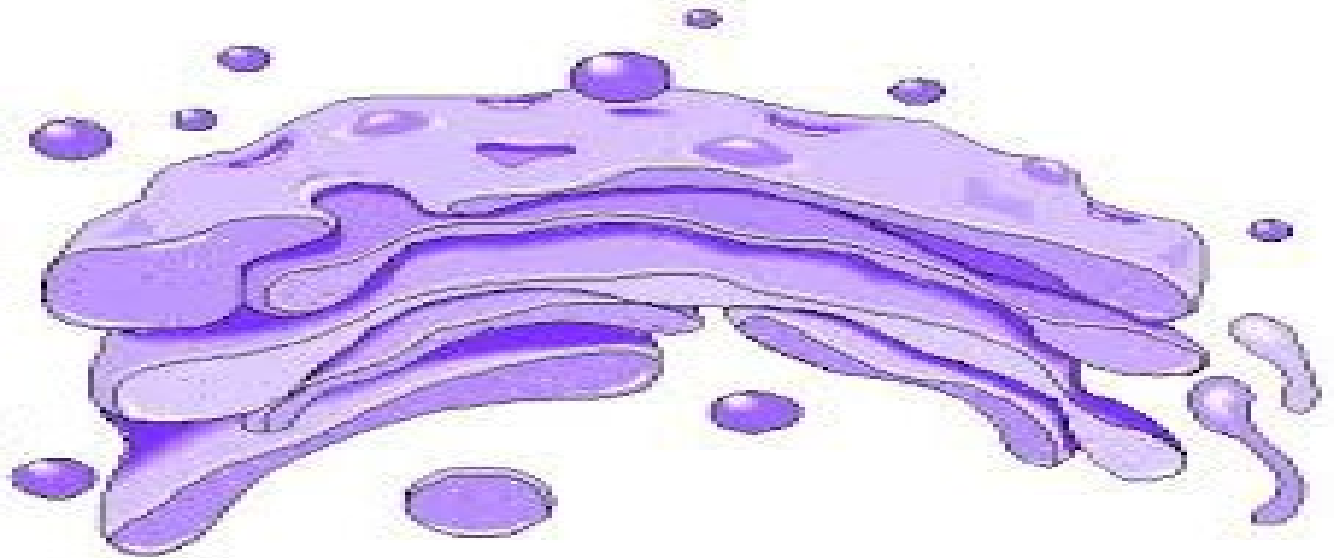


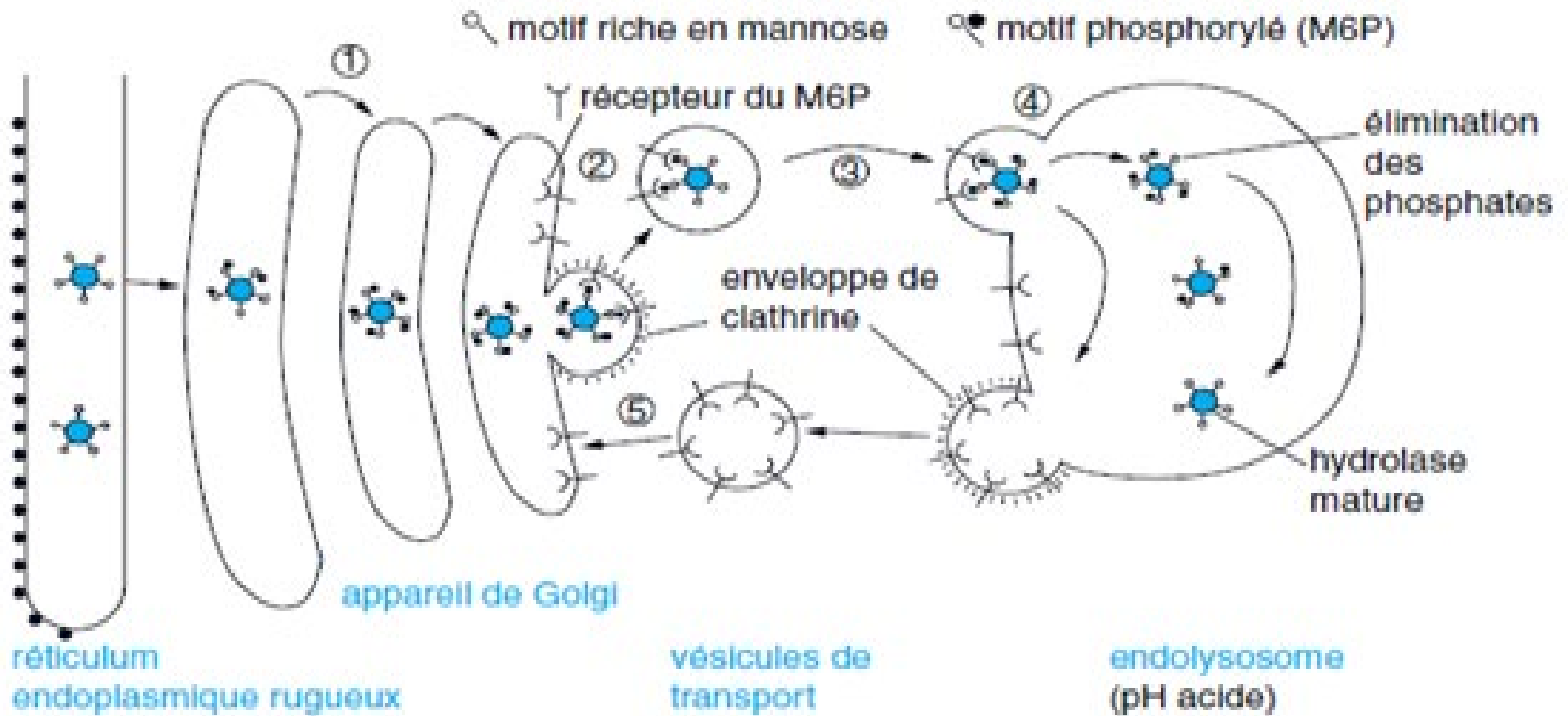
An external face or cis face in the vicinity of the ER. At this level there is budding of a specialized region of the ER to give transitional vesicles. The latter will pass through the cis Golgian network and merge with a membrane type according to their content. An inner or trans side formed by more dilated saccules. At its periphery is a trans-Golgian network through which the molecules exit either to the lysosomes or to the secretory vesicles or to the cell surface

La O-glycosylation

Takes place in the golgi apparatus, more precisely in the median and trans saccules. In this glycosylation, the glycosylated chain is transferred to the oxygen carried by the amino acid serine or threonine of the protein by an O-glycosyltransferase.

Unlike N-glycosylation, the oligosaccharide chain is highly variable and is not necessarily bound by an N-acetyl glucosamine group.





Labeling of hydrolases in the Golgi apparatus and addressing to lysosomes

The main steps are: (1) the marking of the mannose with a phosphate group; (2) the binding of the protein to the 6-P mannose receptor; (3) transport to an endosome; (4) dissociation of the protein from its receptor; (5) the recycling of the receptors released to the transgolgi saccules.

It occurs in the Cis saccules of the Golgi apparatus and takes place in several stages:

- Marking of mannoses by phosphotransferases (by adding a phosphate group on the 6th carbon of one or more mannoses).

Once labelled, the enzymes will no longer be able to undergo enzymatic changes in the rest of the golgi.

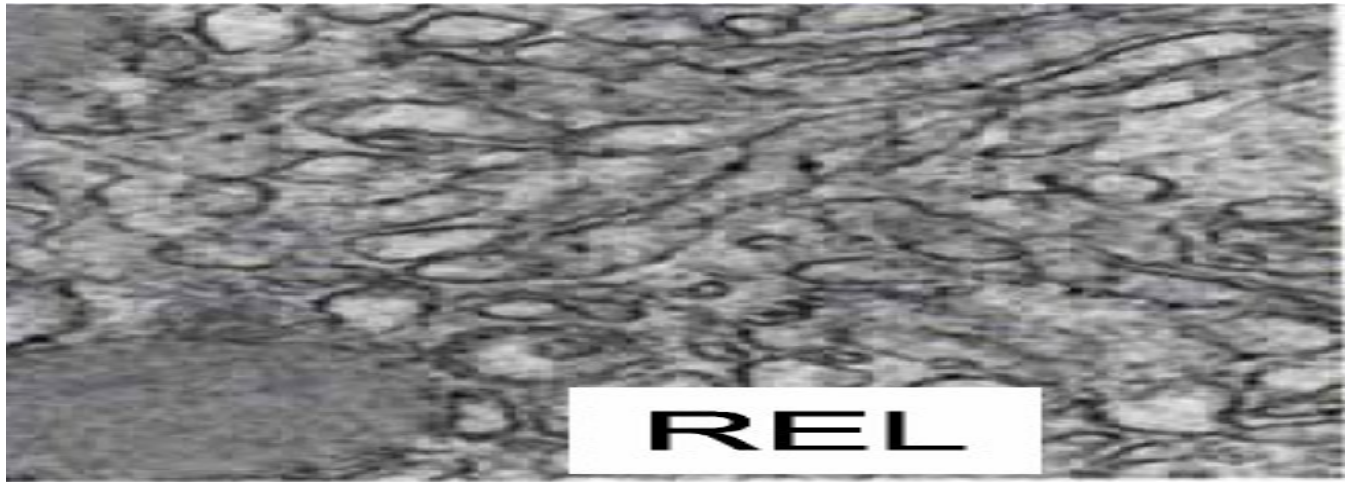
- Enzymes carrying 6P mannose are recognized by their specific receptors at the TGN level.
- Formation of clathrin-coated vesicles that bud from TGN.
- Removal of clathrin coating from the vesicle after budding.
- Fusion of the vesicles with the endolysosome.

Under the effect of the acidic pH of the endolysosome, the protein detaches from its receptor:

- The enzymes become mature after removal of the phosphate group by phosphatases.
- Free 6P mannose receptors are recycled to the Golgi apparatus (clathrin vesicle).

Role of the Smooth Endoplasmic Reticulum in lipid synthesis

“Self-fabricating or biogenic membrane”



- Assembly of lipid bilayers in the ER with synthesis of all major classes of lipids, phospholipids and cholesterol on the cytosolic side.
- Source of lipids for the outer membrane of the mitochondria.

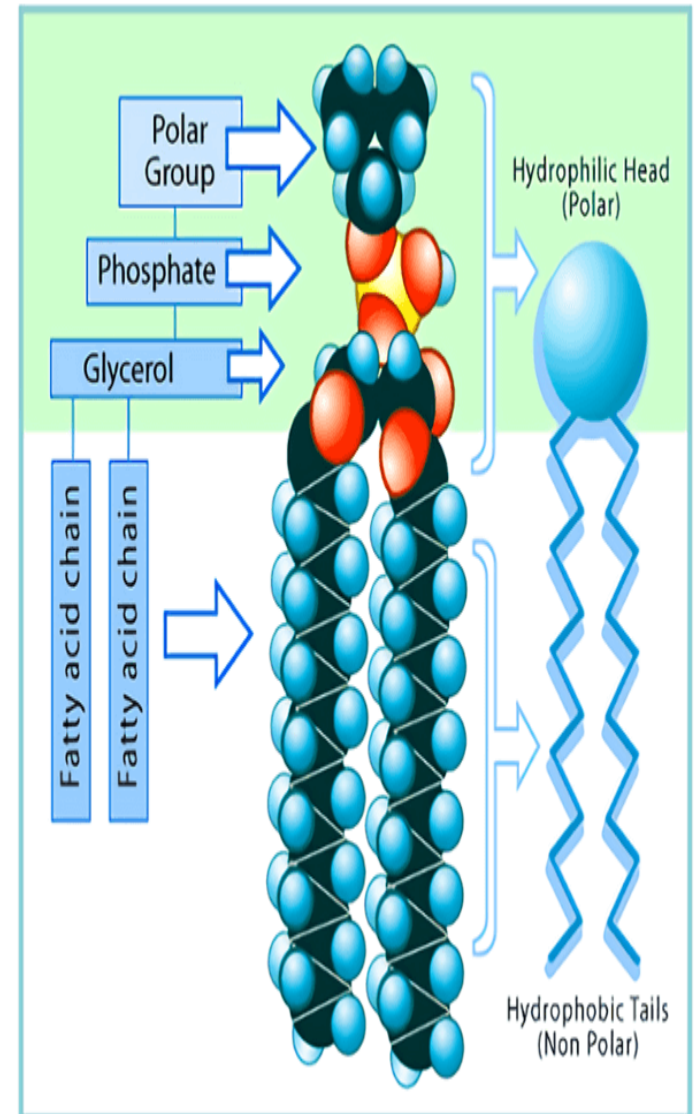
Synthesis of Membrane Phospholipids

Phospholipid synthesis occurs exclusively in the cytosolic leaflet of the smooth endoplasmic reticulum (SER).

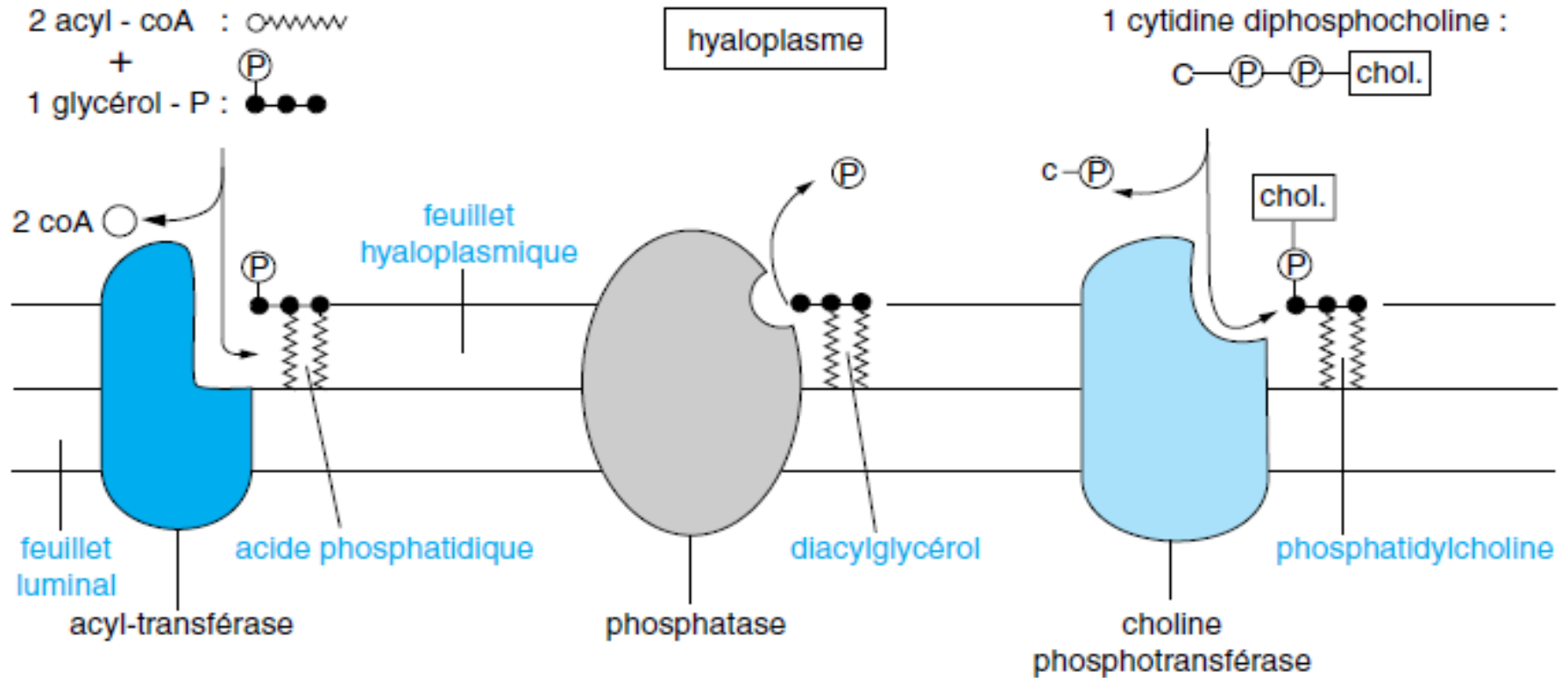
Example: Phosphatidylcholine synthesis

All the enzymes involved in this synthesis are proteins contained within the SER membrane, whose active sites face the cytosol. Indeed, all the substrates involved in the production of this lipid originate from the cytosol:

- fatty acids (in the form of acyl-CoA: the soluble form of these molecules).
- glycerol (in the form of glycerol phosphate).
- and choline (in the form of CDP-choline).

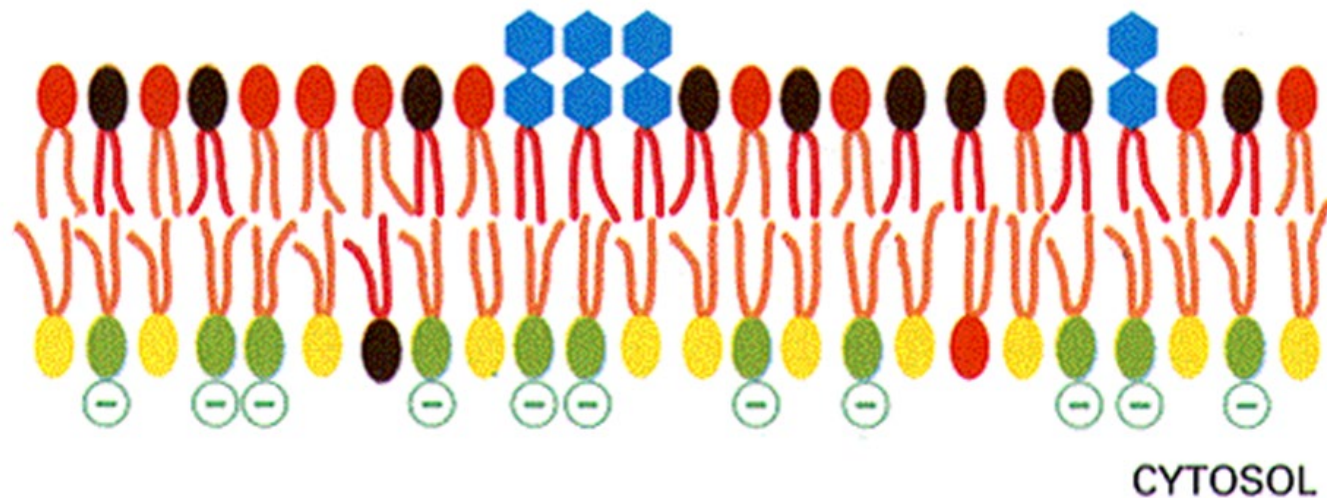


Example: Phosphatidylcholine Synthesis



1. Two acyl-CoAs are first successively attached to a glycerol phosphate by acyltransferase while releasing CoA (acylation): obtaining an acid phosphatidic.
2. A phosphatase dephosphorylates phosphatidic acid: obtaining a diacylglycerol (DAG).
3. Addition of phosphocholine to DAG by choline-phosphotransferase.... obtaining phosphatidylcholine.

Phosphatidylserine, phosphatidylethanolamine and phosphatidylinositol are synthesized in the same way as phosphatidylcholine: it is only the last enzyme that differs (addition of the polar head: serine, ethanolamine or inositol).

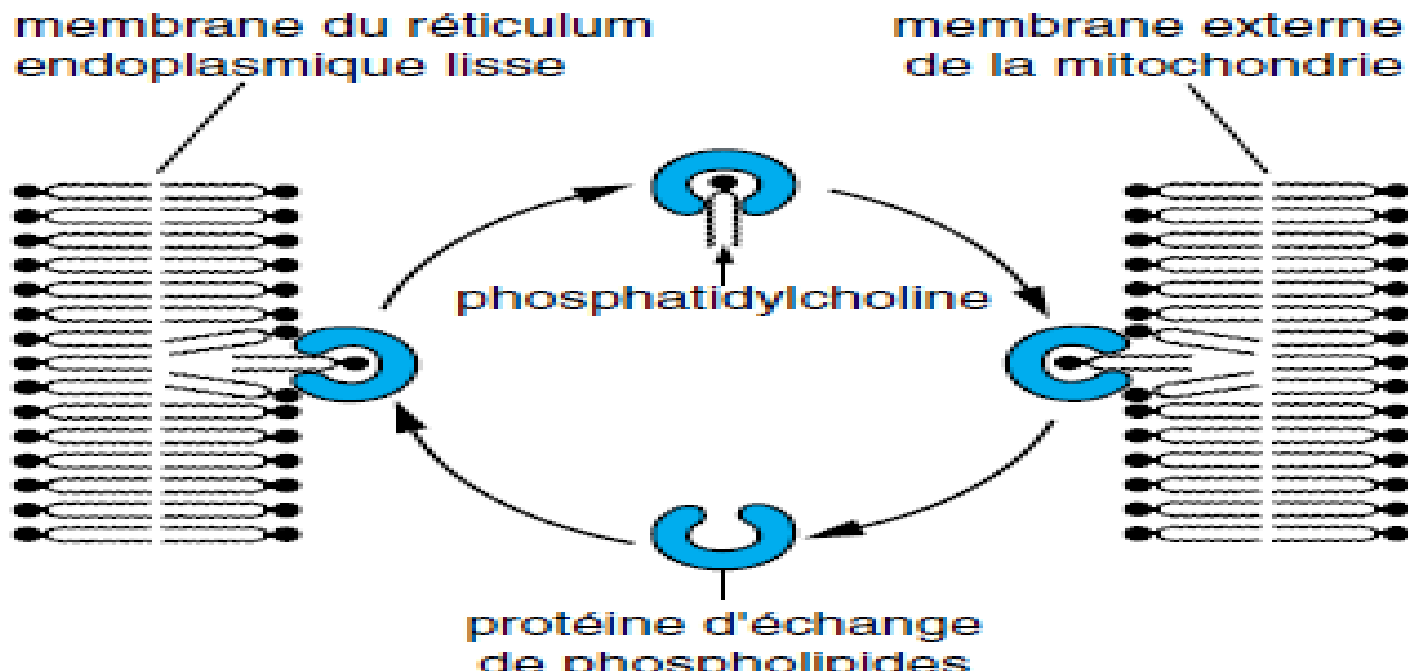


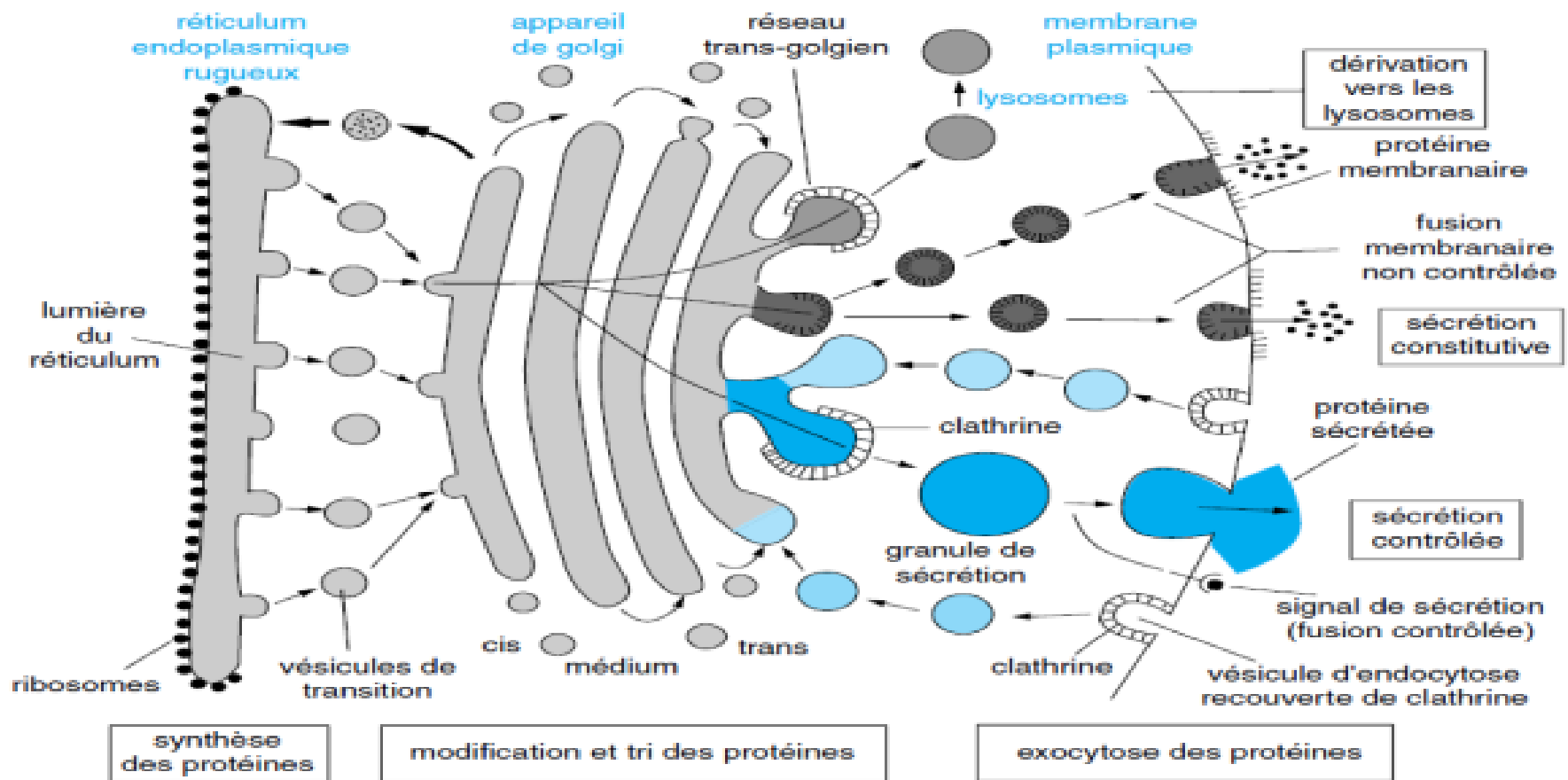
Belonging to the Endomembrane System

The Golgi apparatus, endosomes, lysosomes, and the plasma membrane receive new membrane phospholipids (synthesized in the smooth endoplasmic reticulum) via vesicular transport.

The transfer of phospholipids to compartments outside the endomembrane system is mediated by soluble carrier proteins in the cytosol.

These proteins are called phospholipid exchange proteins (or PL transfer proteins).





Various pathways for the survival of proteins through the Golgi apparatus

Three pathways are identified, which transport proteins to lysosomes, the plasma membrane, or the extracellular environment.

Two modes of secretion must be distinguished: **the controlled mode**, which concerns specialized secretory cells, and the **constitutive mode**, which concerns all cells.

The retrograde transport of proteins from the endoplasmic reticulum (resident proteins and associated receptors) is mentioned.

Note the presence or absence of Clathrin on some of the vesicles involved in these processes.

Controlled secretion: also called discontinuous secretion, it is a selective secretion regulated by intracellular or extracellular signals and involves only secretory cells. Examples include the secretion of hormones, digestive enzymes, etc.

It should also be noted that the product destined for lysosomes and controlled secretion is often packaged in clathrin-coated transport vesicles.

Constitutive secretion: is a secretion that occurs continuously and without the intervention of signals. It involves almost all cell types. Examples include the renewal of membrane components and the renewal of extracellular matrix components.

The product destined for constitutive secretion is packaged in coatomere-coated vesicles (COPIs).

Anterograde transport: ER to AG – COP2

Retrograde transport: AG to ER – COP1

The Endosomes

Endosomes are membrane compartments found inside eukaryotic cells. It is an organelle of the endocytic membrane transport pathway from the trans-Golgian network:

The endocytosis vesicles attach to them and fuse to release their contents. Internalized molecules or ligands of the plasma membrane (contained in these vesicles) can follow this pathway to the lysosomes to be degraded.

Molecules are also transported to endosomes from the trans-Golgi network and either continue to lysosomes or are recycled back to the Golgi apparatus.

Endosomes can be classified into 2 categories: early or late, depending on their stage of post-internalization.

Early endosome

Early endosomes are organelles located at the periphery of the cell, they receive macromolecules and solutes from the extracellular environment. The primary function of early endosomes is to sort these cargoes into the recycling and degradation compartments of the cell.

They have a pH equal to 6, because in their membranes they have an ATPase proton pump that transports H^+ hydrogen ions from the cytosol to the inside of the endosomes, which causes the dissociation of most of the endocytosis ligands still attached to their receptors. .

Once in the early endosome, the endocytosis material can follow two pathways:
Be recycled by means of vesicle recycling and return to the membrane domain.
Follow the path of the late endosome to be degraded.

Late endosome

The pH has now dropped to 5.0 and the hydrolase content is very high.

Late endosomes are found in the juxtanuclear regions and concentrated at the organizing center of the microtubules. They are an important sorting station in the endocytic tract.

Recycling to the plasma membrane and to the Golgi apparatus occurs in late endosomes.